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Mini Project Abstract Presentation

Lora Based Multihop Predictive Energy Demand Modelling For Wireless Communication Networks

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Abstract:-

- This project proposes a LoRa-based multihop wireless communication system integrated with predictive energy demand modelling. Machine learning algorithms are used to predict the future energy demand of wireless sensor nodes. Based on this prediction, the system can support energy-aware routing decisions. The proposed approach helps improve network lifetime, scalability, reliability, and overall energy efficiency in wireless communication networks.
- LoRa (Long Range) communication is widely used in low-power wide-area networks (LPWAN) for IoT applications. The system enhances energy efficiency in wireless sensor networks (WSNs). Predictive algorithms estimate future energy requirements to enable optimized routing and transmission.

Problem Statement:-

- Traditional wireless sensor networks suffer from high energy consumption, limited communication range, and poor scalability.
- In single-hop LoRa communication, nodes far from the gateway consume excessive power due to long-distance transmission.
- Existing systems do not predict future energy demand, which leads to inefficient routing and reduced network lifetime.
- Therefore, there is a need for predictive and energy-aware multihop communication.

Literature Survey:-

- From literature review, we found that most existing LoRa systems use single-hop communication.
- Traditional routing protocols are static and do not consider predictive energy optimization.
- Several IoT research works use machine learning for forecasting and optimization.
- However, limited work combines LoRa multihop communication with predictive machine learning models for energy-aware routing.

Objectives:-

->The objectives of our project are:

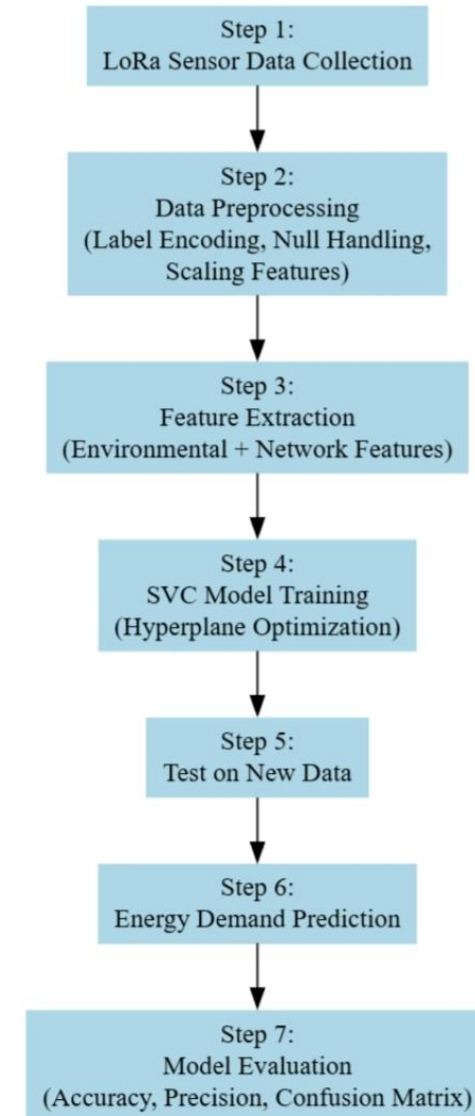
1. First, to develop a LoRa multihop communication framework.
2. Second, to predict future energy demand of LoRa sensor nodes.
3. Third, to optimize routing based on predicted energy requirements.
4. Finally, to improve overall network lifetime and scalability.

Existing Work:-

- Existing systems mainly use single-hop LoRa communication.
- Routing decisions are static and based only on current node parameters.
- Energy optimization is reactive rather than predictive.
- As a result, these systems experience inefficient routing and faster battery depletion.

Proposed Methodology:-

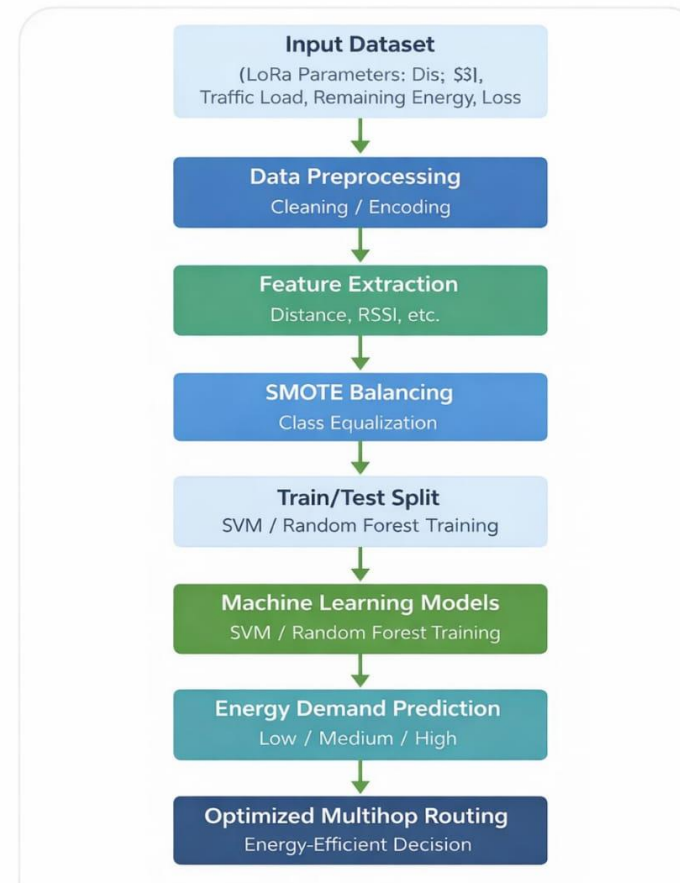
- In our proposed system, we first collect network parameters such as distance, remaining energy, traffic load, RSSI, and packet loss.
- The dataset is preprocessed and balanced using SMOTE.
- We then train machine learning models including Support Vector Machine and Random Forest.
- These models predict the future energy demand of each node.
- Based on predicted energy demand, routing decisions can be optimized for energy-efficient communication.



Block Diagram:-

- This diagram illustrates the workflow of the proposed system, where LoRa network data is preprocessed and analyzed using machine learning models to predict energy demand. Based on the prediction, optimized multihop routing is performed for efficient wireless communication.

LoRa Based Multihop Predictive Energy Demand Modelling



System Design:-

- The system consists of multiple LoRa sensor nodes connected in a multihop network.
- Sensor nodes communicate through intermediate relay nodes.
- A gateway collects network data.
- The machine learning prediction engine processes this data and estimates future energy demand.
- Routing can then be optimized based on these predictions.

Hardware Deatails:-

- The hardware requirements for this project include a laptop or PC for implementation.
- Optionally, LoRa modules and wireless sensor nodes can be used for prototype deployment.
- In our project, the implementation is simulation/software-based.

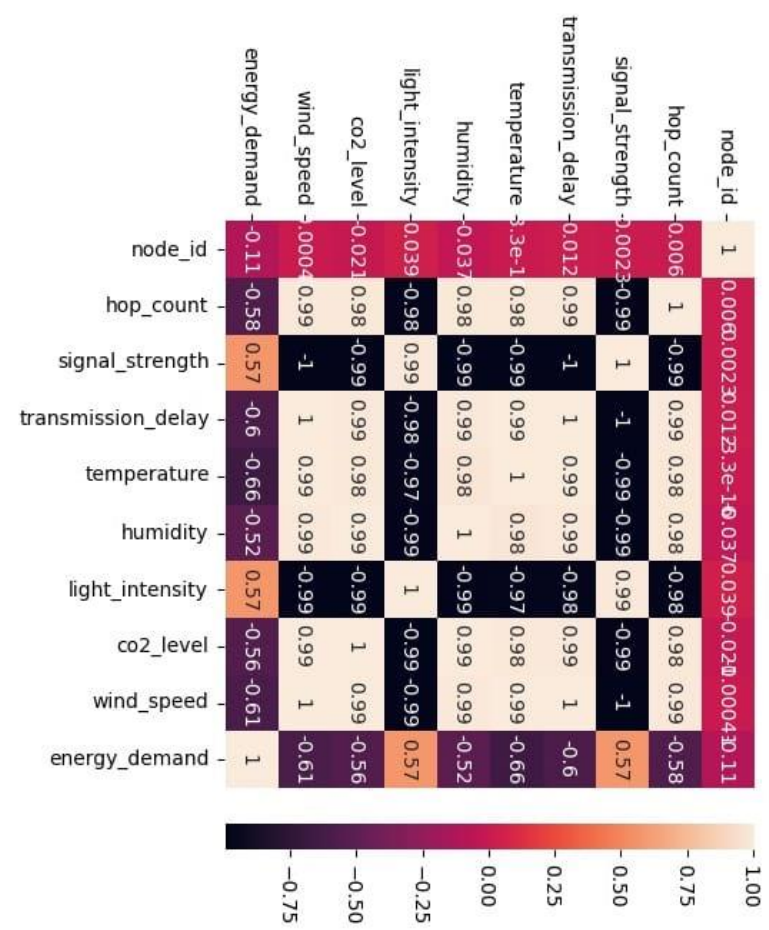
Software Details:-

- The software tools used are Python 3.x and Visual Studio Code.
- We used NumPy and Pandas for data handling,
- Scikit-learn for machine learning model implementation,
- and Matplotlib/Seaborn for visualization.

Implementation Flow:-

- First, the training dataset is loaded into Python.
- Then labels are encoded and the data is split into training and testing sets.
- SMOTE is applied for balancing class distribution.
- SVM and Random Forest models are trained.
- Performance metrics are calculated.
- Finally, test data is passed for energy demand prediction.

Result:-



Predictions:					
	Node_ID	Distance_m	Remaining_Energy_%	Traffic_Load_pkts	\
0	1	120	95	20	
1	2	250	88	35	
2	3	180	90	25	
3	4	300	80	40	
4	5	220	85	30	

	Packet_Rate_pkts_per_min	RSSI_dBm	Packet_Loss_%	Predicted	Energy Demand
0	15	-85	1.2		Low
1	18	-92	2.5		Medium
2	12	-88	1.8		Medium
3	20	-95	3.1		High
4	16	-90	2.0		Medium

Applications&Future Scope:-

->Applications

- Smart Agriculture
- Smart Cities
- Industrial IoT

->Future Scope

- Real-Time Edge Deployment
- Advanced Deep learning models
- Adaptive energy routing

Reference:-

- (IEEE)
- F. Adelantado et al., IEEE, 2017
- A. Augustin et al., Sensors Journal, 2016
- LoRa Alliance Documentation
- Arduino & MATLAB Documentation